

Physiology of the digestive system



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WPS Office تعديل من خلال



Over all function(s) of the GIT

- ❖ To provide the organism with the necessary ingredients (food) to maintain life and vitality
- ❖ To rid the body of toxic materials in the form of faeces



From basic physiology

Transport across the cell membrane will come into play when we discuss

- Absorption
- Secretion
- Reabsorption

For transport to take place, we need
Permeability and Driving force (pressure gradient or concentration gradient)



From biochemistry

The whole metabolism is either

- Catabolism---degrading stuff
- Anabolism---building stuff



Processes of the GIT

➤ Digestion:

- Is the process of breaking down food into smaller manageable particles

Mechanical and chemical

➤ Absorption:

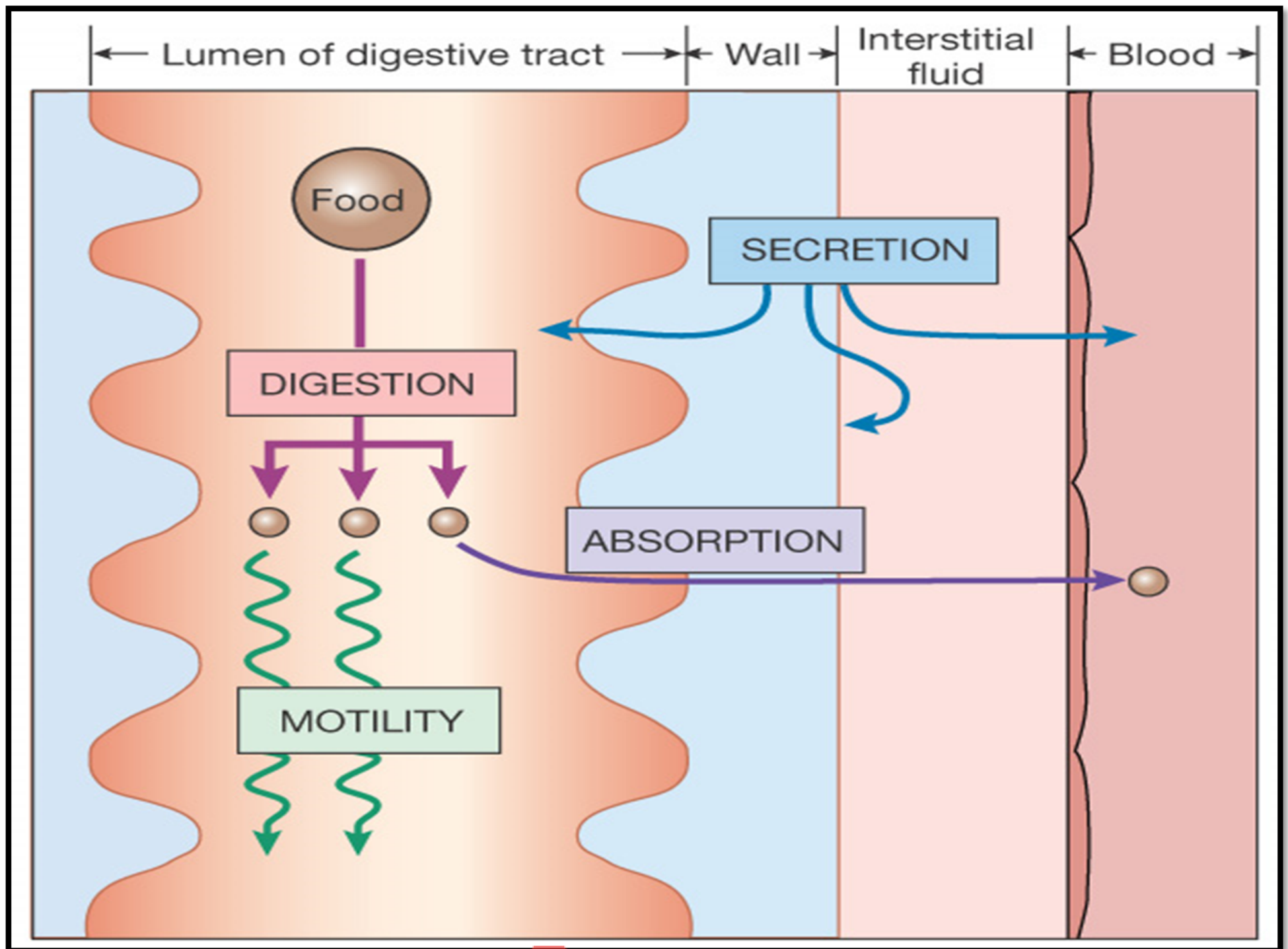
- Transport across a bio-membrane from the lumen into the cells, ISF or blood



Processes of the GIT

- Reabsorption: Transport across a bio-membrane of a substance that has been absorbed before, from the lumen back into cell, ISF or blood
- Secretion: Transport across a bio-membrane of a substance from cell, ISF or blood into the lumen





The anatomical composition

➤ Organs:-

- Mouth (oral cavity), pharynx, esophagus, stomach, small intestine & large intestine.

➤ Associated organs:-

- Salivary glands, pancreas & liver and others...



The oral cavity

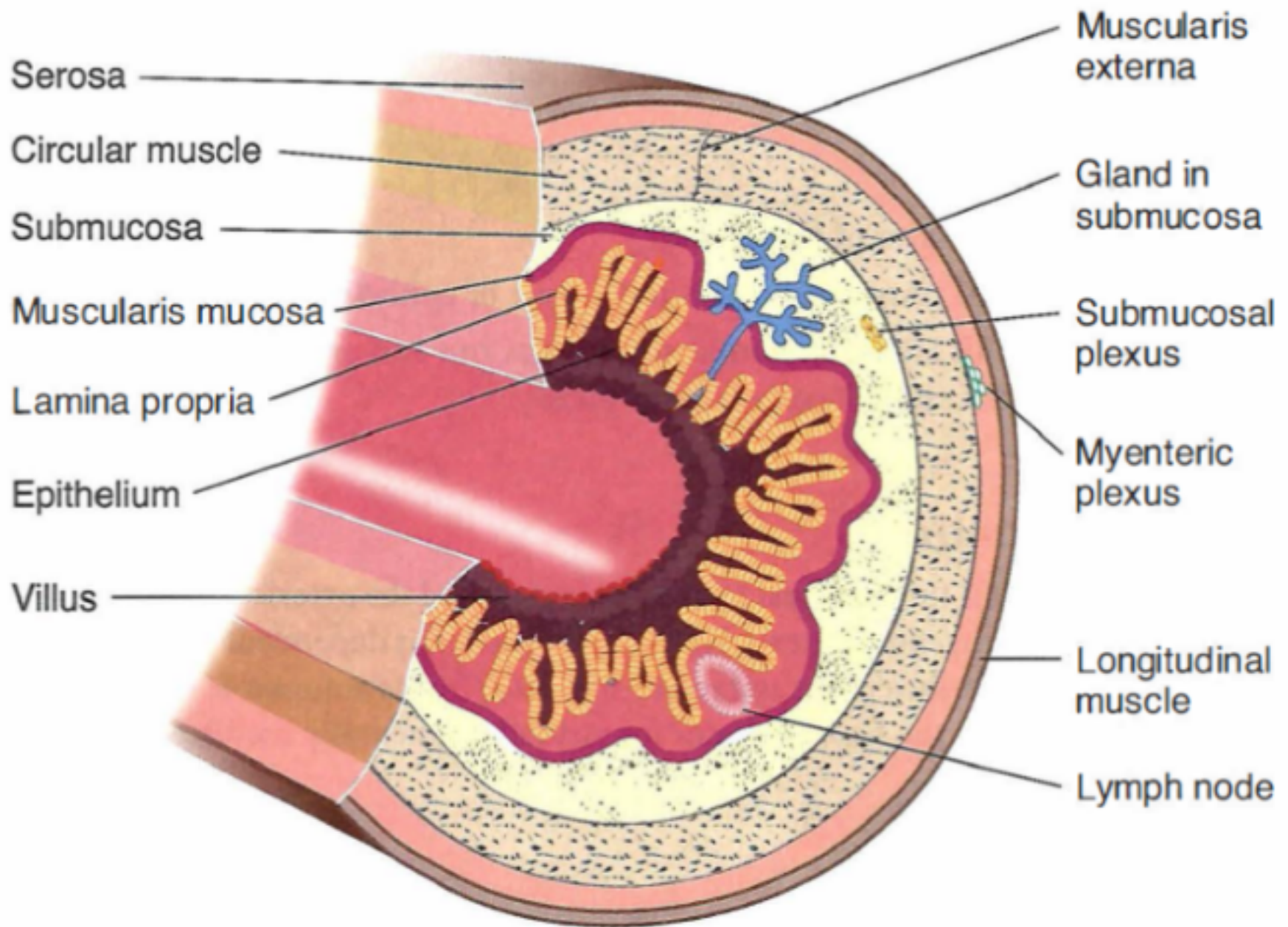
➤ Functions:-

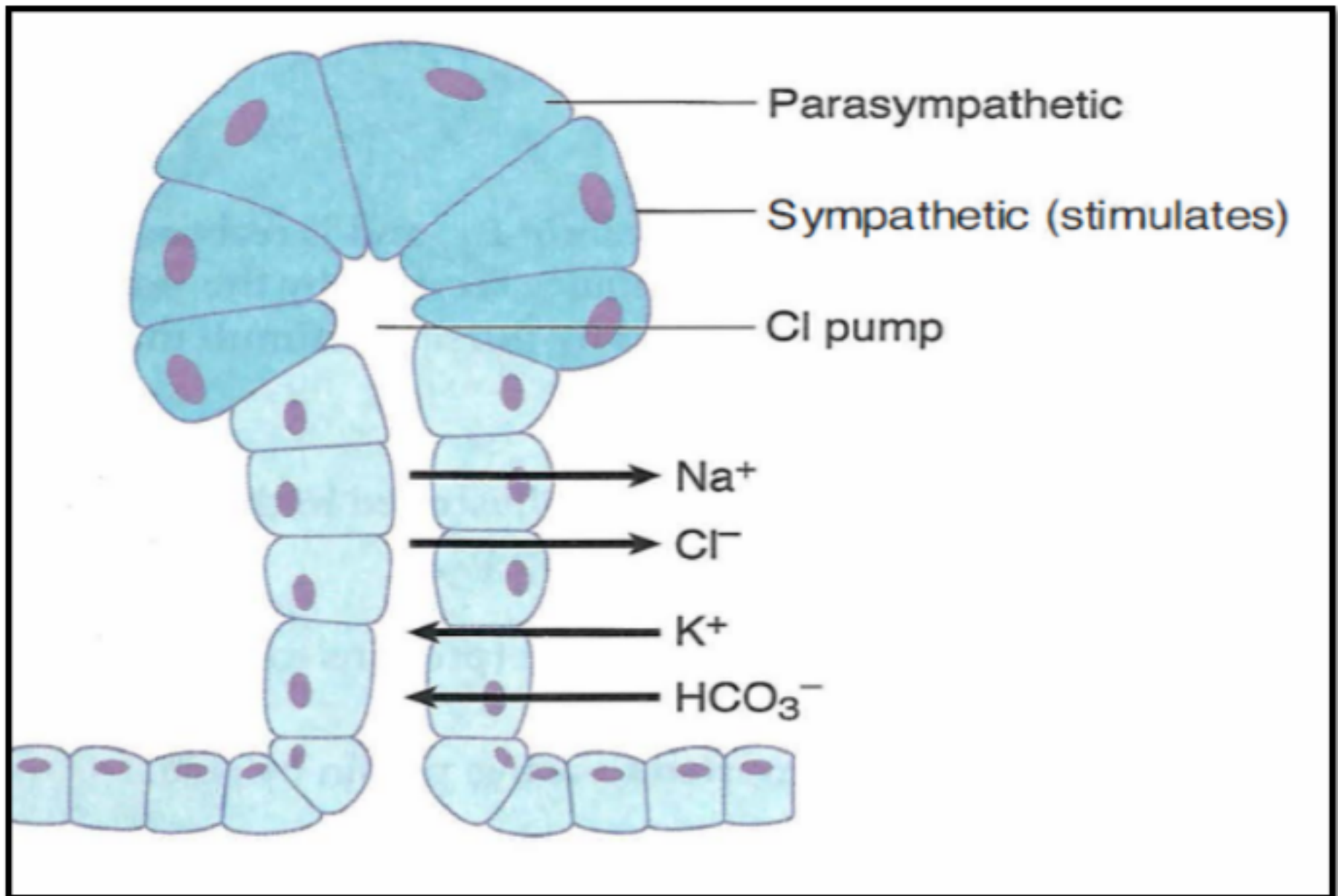
- Mastication (the mechanical breakdown of food).
- Mixes food with saliva.
- Initiates swallowing.

➤ Saliva:-

- Fluid poured into the oral cavity produced by salivary glands.



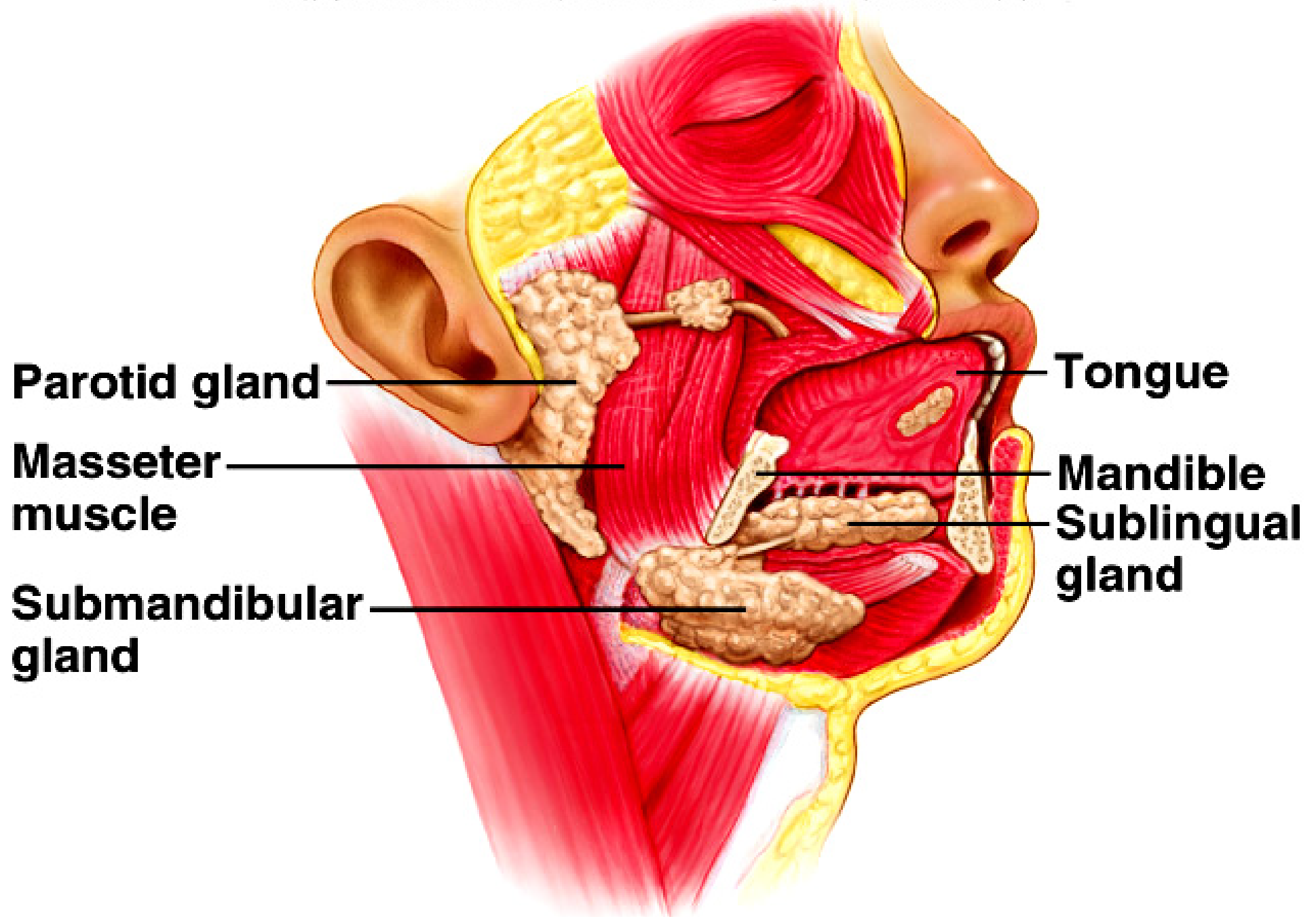




➤ The salivary glands include:

- The parotid glands
- The submandibular gland
- The sublingual gland





Functions of the saliva

- 1-Protection:** by lysozome & antibodies.
- 2- Digestion:** by alpha amylase & lingual lipase.
- 3-Lubrication to facilitate:** swallowing, speech & taste.
- 4-Buffer:** neutralizes acids & relives heartburn.



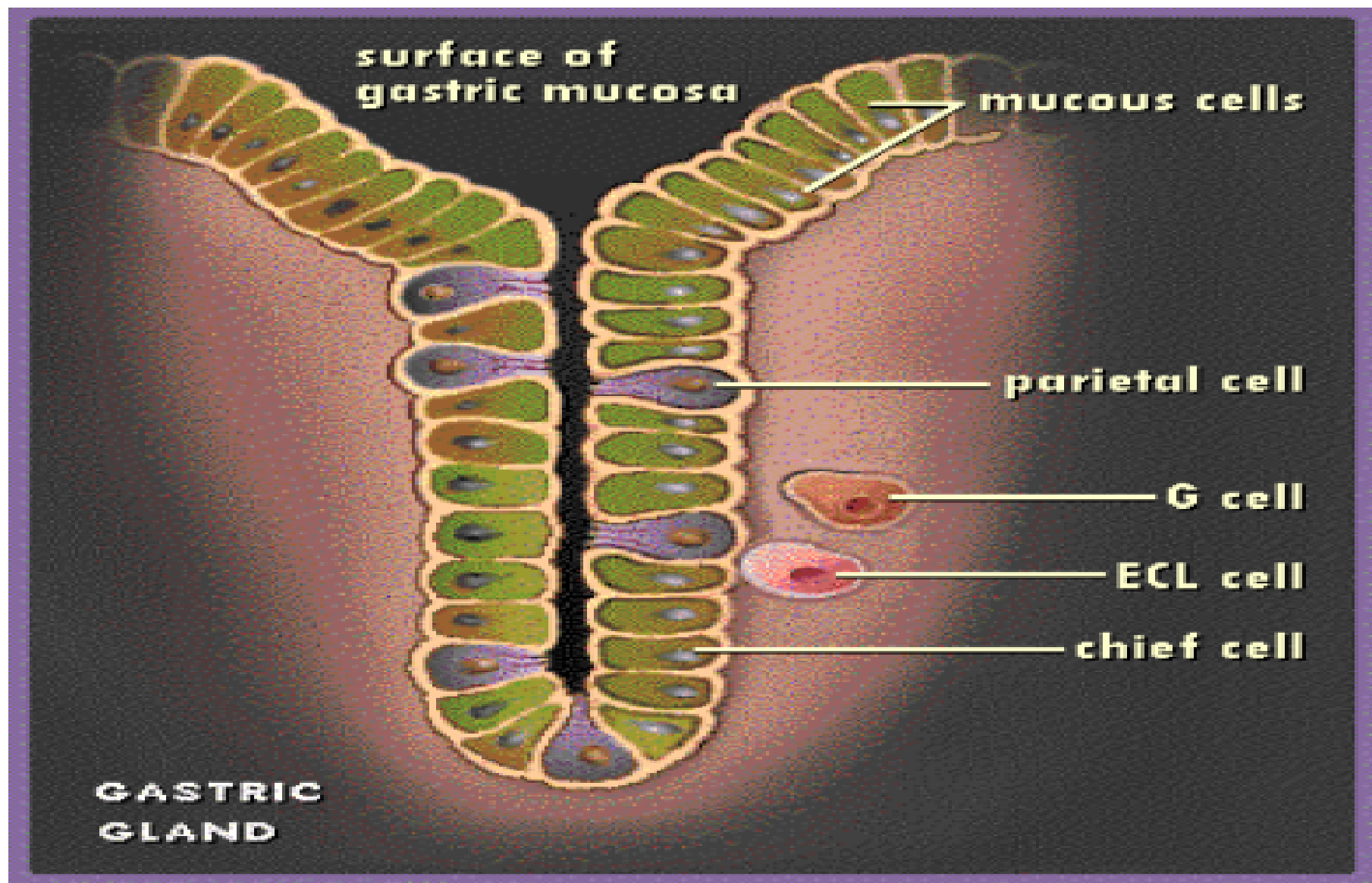
Stomach



- The mucosa of the stomach contains many exocrine glands (gastric glands).

➤ Types of cells in a gastric glands:

- 1- Mucous cells: secretes mucous.
- 2- Parietal cells: secretes HCL & intrinsic factor.
- 3- Chief cells: secretes pepsin enzyme for protein digestion.



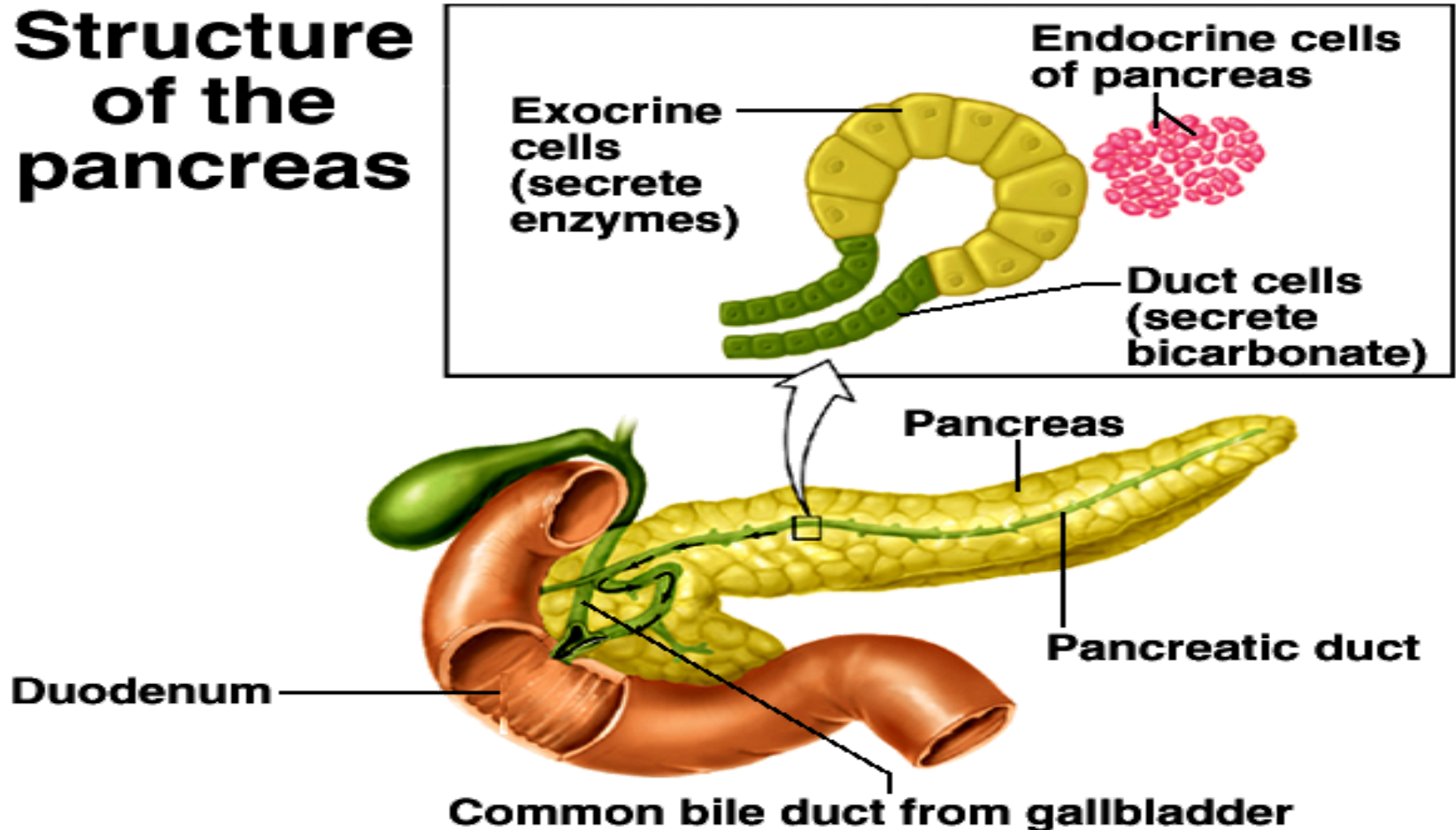
Functions of the stomach

1. Stores food up to about 4 hours, and controls its release into the duodenum.
2. Digestion of proteins & lipids start in the stomach.
3. Protection by HCL.
4. HCL also maintains iron in the ferrous state to facilitate its absorption.
5. Production of intrinsic factor to facilitate absorption of vitamin B12.
6. Hormonal function: produces the hormone Gastrin to increase HCL production



THE PANCREAS

Structure of the pancreas



The Pancreatic Enzymes

- The main digestive enzymes in the body.
- For digestion of all types of food (protein, carbohydrate & lipids).

1-Trypsin: for proteins digestion :

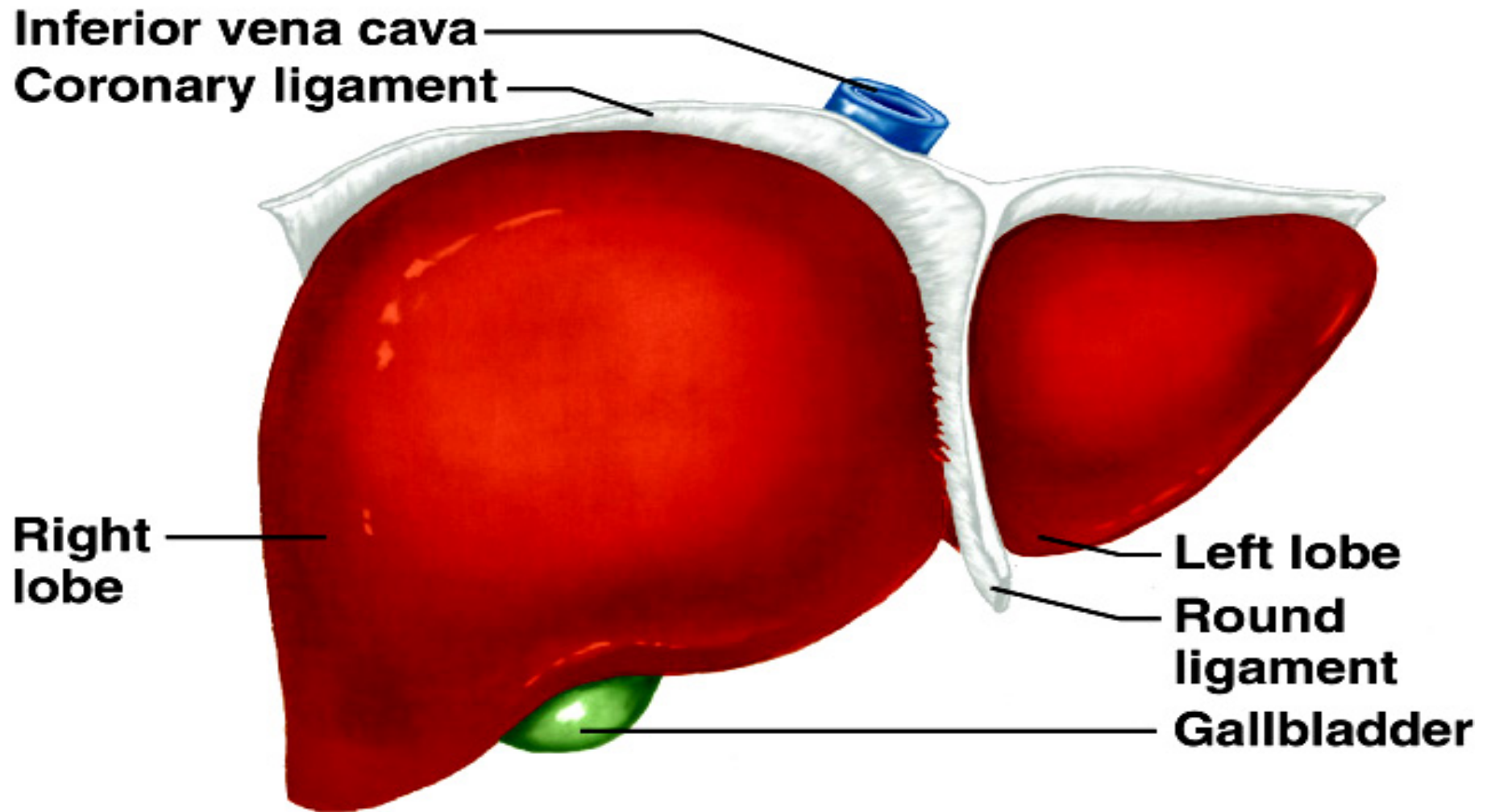
2- Pancreatic alpha amylase: for carbohydrates digestion .

3- Pancreatic lipase: for lipid digestion



The Liver

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➤ Functions of the liver:

1. Synthesis of the plasma proteins.
2. Synthesis and secretion of bile.
3. Metabolism of CHO, proteins, fats & some hormones.
4. Excretion of end products of metabolism through the bile.
5. Storage of glycogen, iron and vitamins (A,D & B12).
6. Detoxification.



BILE

- Yellowish- greenish fluid secreted continuously by hepatocytes into the biliary system.

➤ Functions:

- Neutralization of acid in the duodenum by the alkaline pH.
- Fat digestion & absorption (by bile salts).
- Excretion of bile pigment (bilirubin), cholesterol, drugs and dyes



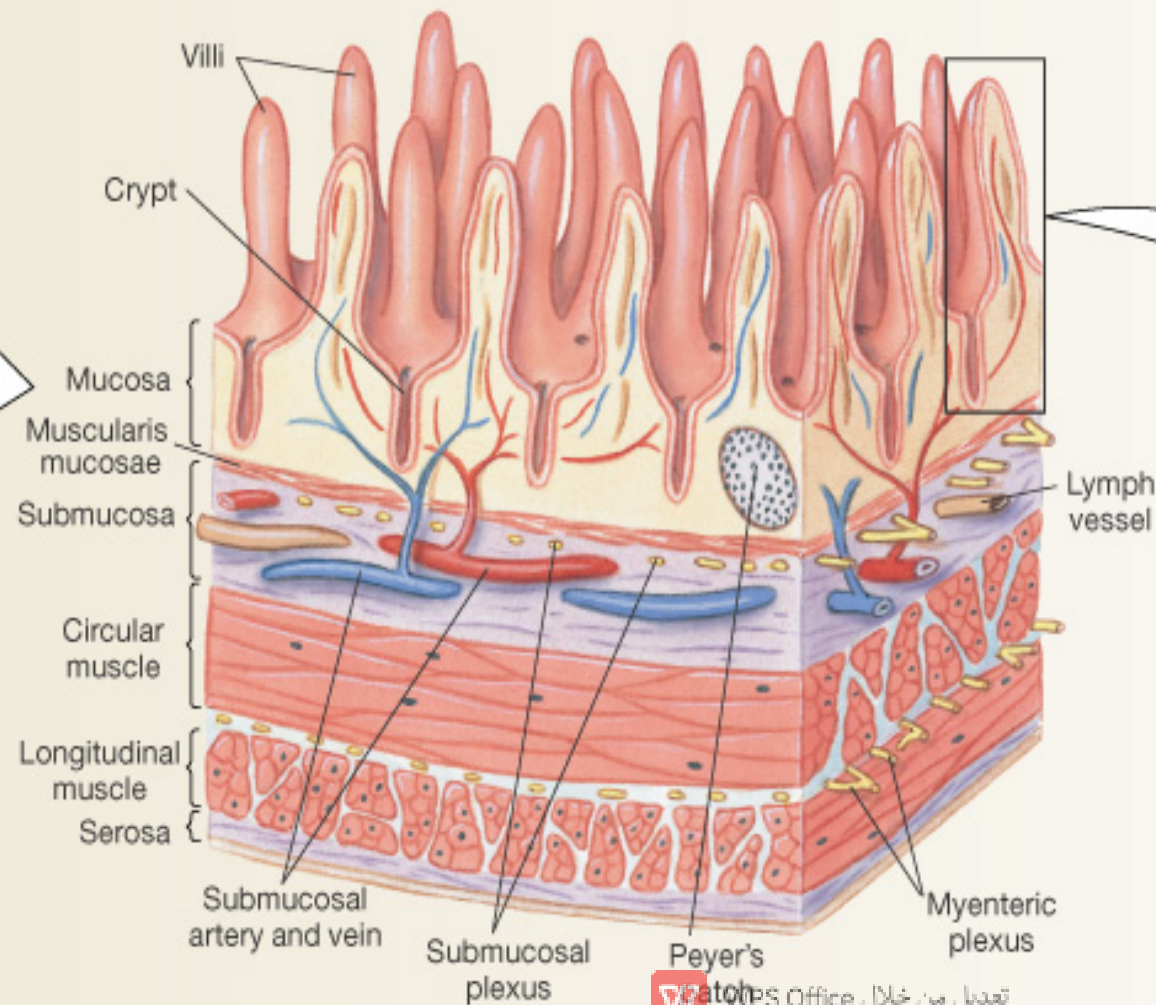
The Small Intestine

- Small intestine is essential to life because it is the most important site for digestion & absorption.
- Enzymes: Enteropeptidase, disaccharidases (maltase, sucrase & lactase)

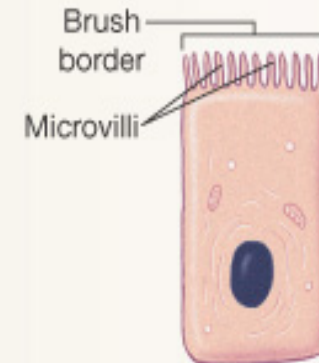


SECTIONAL VIEWS OF THE STOMACH AND INTESTINE

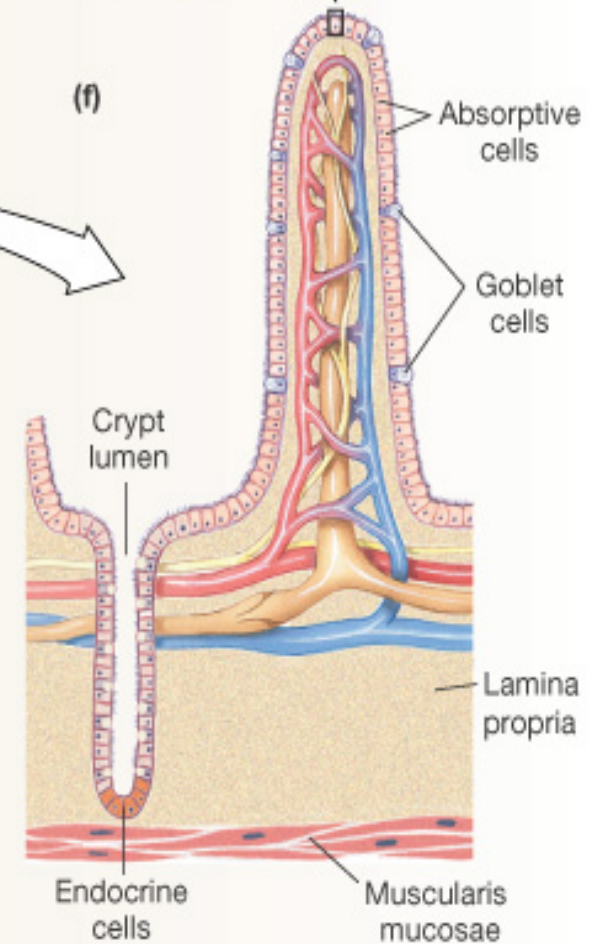
(e) Intestinal surface area is enhanced by finger-like villi.

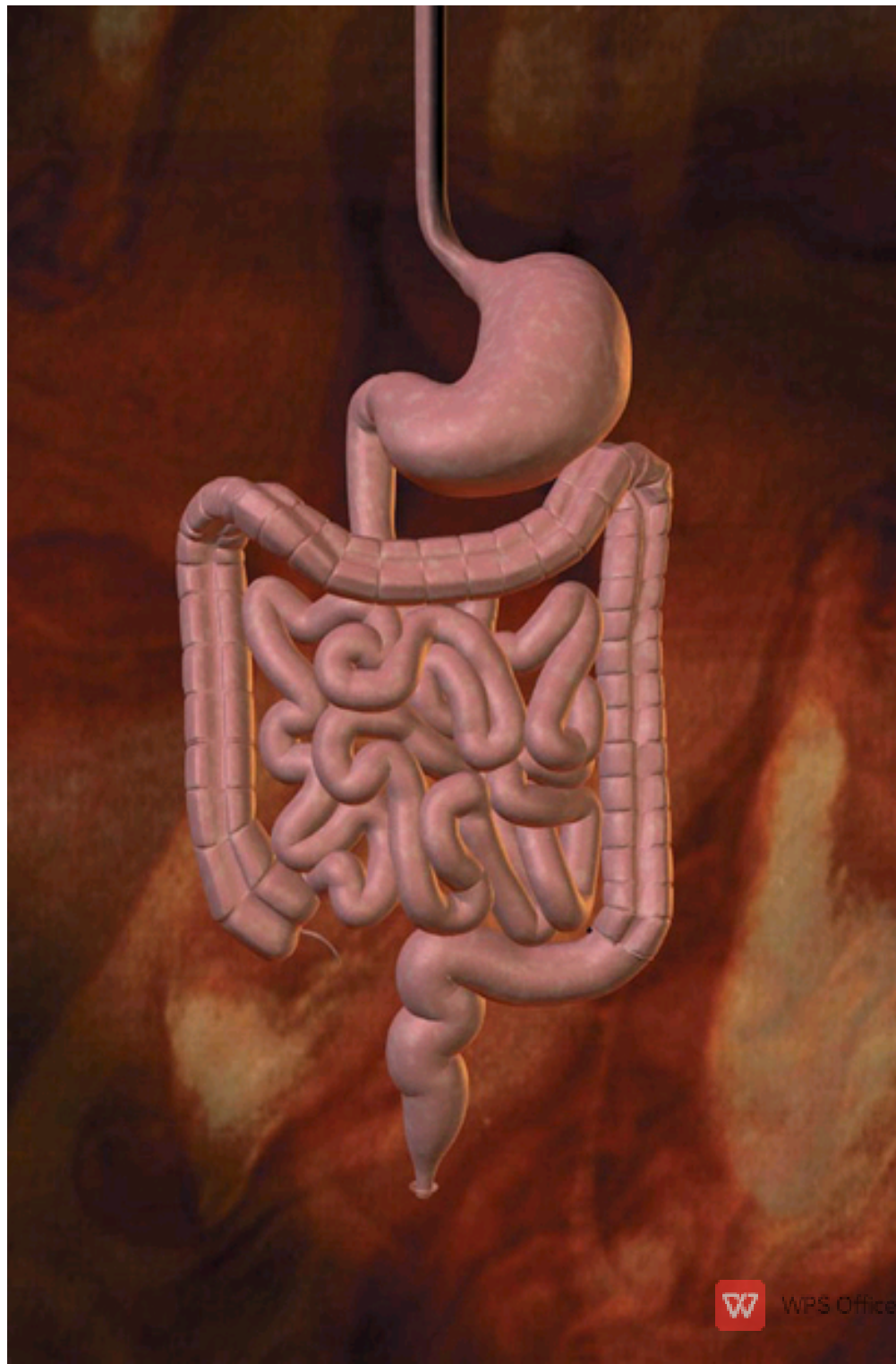


(g) Absorptive cell



(f)





small intestine

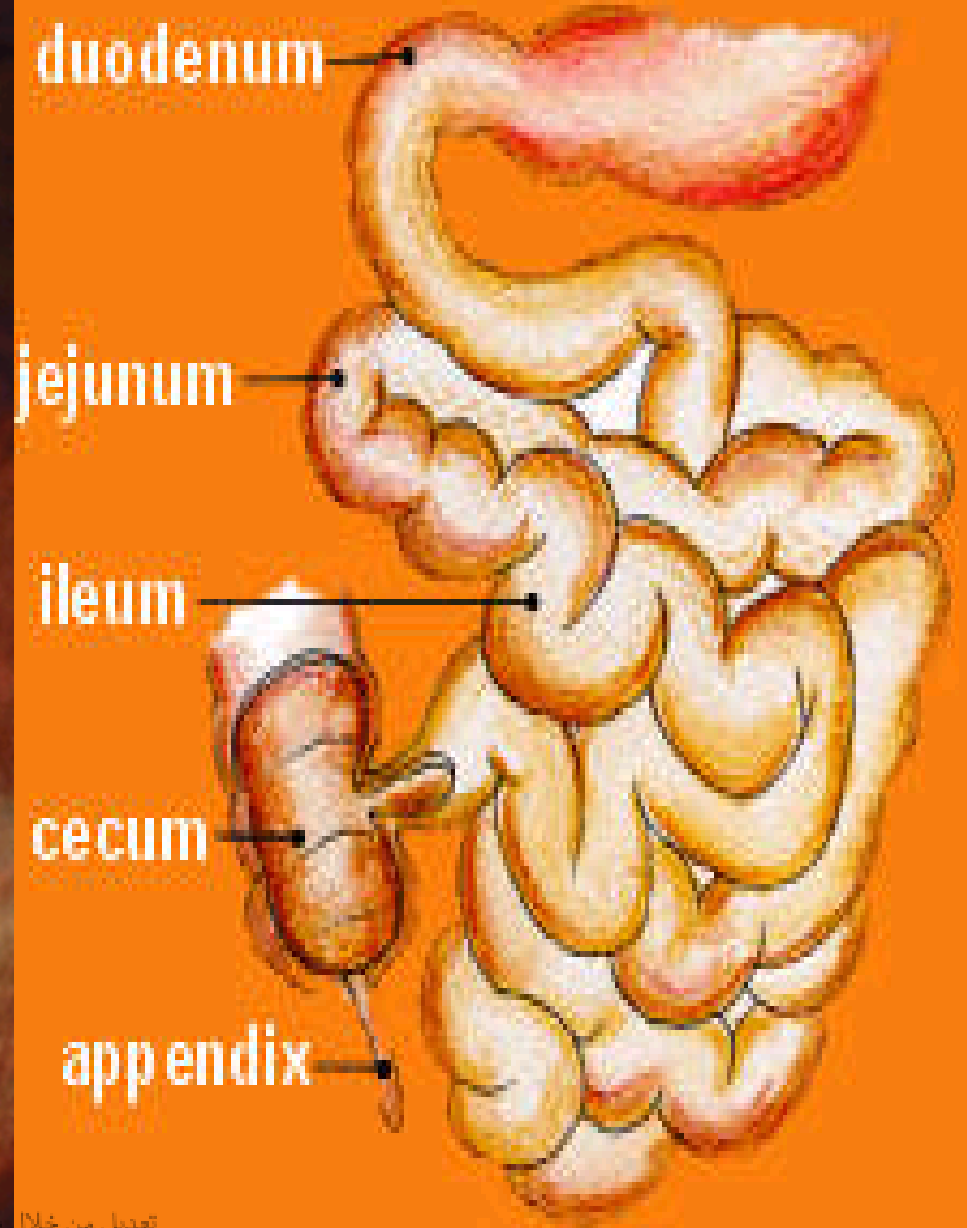
duodenum

jejunum

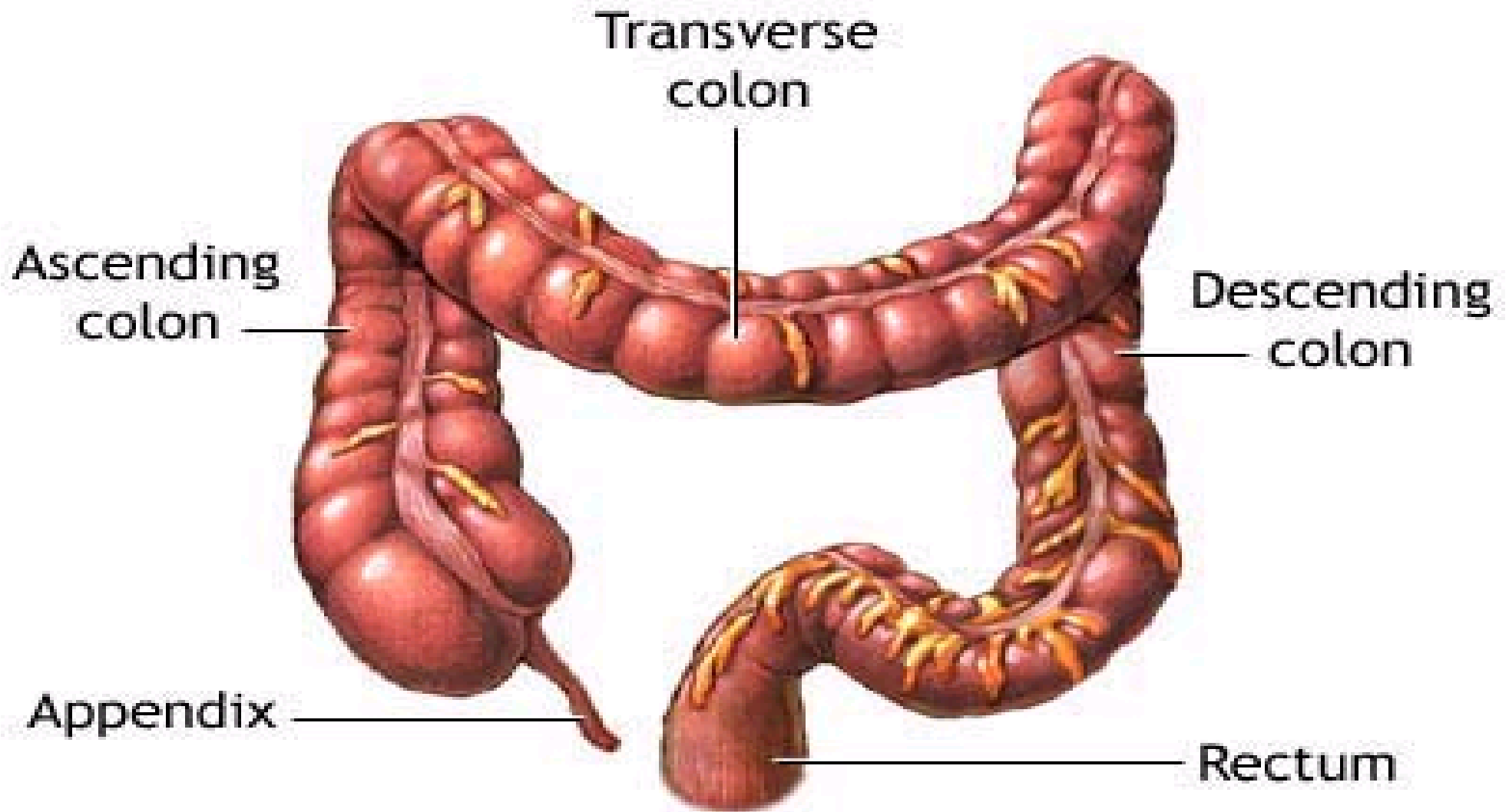
ileum

cecum

appendix



The Large Intestine



Colon

- Reabsorb water from food and digestive juices

